

Rentals KPI Formula Sheet

EstateCFO - Overview, Financials, CFO Dashboard, Financial Ratios, AR Dashboard

Rentals KPI Formula Sheet

Reference for KPI calculations across Rental Overview, Financials KPI Dashboard, CFO Dashboard, CFO Portfolio, and Financial Ratios.

Excel / CSV export (open in Excel)

File	Purpose
docs/Rentals-KPI-Formulas.csv	Full table: Dashboard, Formula, Data Source, API, DB tables, Benchmark
docs/Rentals-KPI-Formulas.pdf	Printable PDF - run python docs/generate_kpi_pdf.py to regenerate

Data sources - which file feeds which dashboard

Input file / action	What it stores	Dashboards that use it
Rent Receivable Excel - Load Portfolio or Sync Rent Data	Unit rent_history, monthly_rent_data, collected, GPR, vacancy	Overview, Company Registry, Companies, CFO Portfolio, AR Dashboard
Financials Excel (P&L + BS + CF) - Rentals -> Financials upload	r_financial_uploads	KPI Dashboard, CFO Dashboard, Financial Ratios
QB AR Aging upload	Aging buckets	Overview DSO, AR Dashboard
Manual expenses	r_expenses	NOI everywhere
Loans module	EMI, balance, LTV	CFO Portfolio DSCR, Financial Ratios

Collected priority (operational): Rent Receivable -> P&L -> invoice collections.

KPI Dashboard / Financial Ratios: Uploaded P&L + Balance Sheet only (rentalKpiEngine.ts).

A) Rental Overview (operational dashboard)

Data source: Rent Receivable Excel (primary), expenses, QB AR Aging, invoice fallback API: GET /api/rentals/portfolio-summary?month=YYYY-MM

Card Label	Formula
Occupancy Rate	Occupied Units / Total Units x 100
Occupied / Vacant	Occupied Units and Vacant Units (vacant = total - occupied)
Collected This Month	Sum of unit rents for selected month from Rent Receivable (monthly_rent_data); else P&L; else invoices
NOI This Month	Collected This Month - Total Expense This Month
Gross Potential Rent	Sum of unit monthly_rent or peak month from Excel
Vacancy Loss	Vacant units' rent or company vacancy_loss from Excel sync
Collection Rate	Collected This Month / Billed This Month x 100
Avg Rent / Unit	Gross Potential Rent / Occupied Units
Arrears Days Outstanding	Weighted DSO from QB aging buckets: (Currentx0 + 1-30x15 + 31-60x45 + 61-90x75 + 90+x105) / Total
Vacant > 30 Days	NA (vacancy date not tracked yet)
NOI Margin	NOI This Month / Collected This Month x 100
Best / Worst (Occ.)	Highest and lowest company occupancy %
Arrears Outstanding	Sum of max(0, Billed - Collected) per invoice
Partner Share Payable	NOI x ownership % for limited/silent partners (only if NOI > 0)

AR section (Overview)

Item	Formula
Arrears Aging buckets	Unpaid amount per invoice aged from billing month due date (1st of month) into Current / 1–30 / 31–60 / 61–90 / 90+
Top Risk Companies	Ranked by Arrears + Vacancy Loss; flagged if arrears > \$2k, occupancy < 85%, or high arrears
Company filter	Uses per-company QB aging totals when uploaded

Backend reference: backend/services/rental_calculations.py (company_summary, arrears_aging, unit_arrears)

B) Financials -> KPI Dashboard (P&L / Balance Sheet)

Data source: uploaded QuickBooks financials (CASH_FLOWS.xlsx) Frontend reference: frontend/src/utis/rentalKpiEngine.ts, frontend/src/pages/RentalFinancials.tsx (KPITab)

Base inputs (from P&L / BS)

Input	Formula
Total Revenue	Total Income row, else sum of income/revenue/rent lines
Total Expenses	Total Expenses row
Net Income	Net Income row
NOI	Net Operating Income row, else Total Revenue - Total Expenses + Interest Expense
Interest Expense	Interest paid lines (absolute value)
Rental Income	Rental income / services lines
Management Fee	Management fee lines
Repairs	Repair / maintenance / cleaning lines
Buildings	Buildings / property & equipment from balance sheet
Long-term Loans	Long-term liabilities / loan accounts
Cash	Total bank accounts
Total Assets / Liabilities / Equity	Balance sheet totals

Profitability KPIs

Card Label	Formula	Target
NOI Margin	$\text{NOI} / \text{Total Revenue} \times 100$	> 40%
Net Income Margin	$\text{Net Income} / \text{Total Revenue} \times 100$	> 10%
Revenue Growth YoY	$(\text{Current Revenue} - \text{Prior Revenue}) / \text{Prior Revenue} \times 100$	> 3%
Expense Ratio	$\text{Total Expenses} / \text{Total Revenue} \times 100$	< 70%

Rental performance KPIs

Card Label	Formula	Target
Rental Income %	$\text{Rental Income} / \text{Total Revenue} \times 100$	> 80%
Interest Coverage	$\text{NOI} / \text{Interest Expense}$	> 2.0x
Mgmt Fee %	$\text{Management Fee} / \text{Total Revenue} \times 100$	< 10%
Repair % of Revenue	$\text{Repairs} / \text{Total Revenue} \times 100$	< 5%

Balance sheet KPIs

Card Label	Formula	Target
LTV (Loans / Building)	$\text{Long-term Loans} / \text{Buildings} \times 100$	< 75%
Asset / Liability Ratio	$\text{Total Assets} / \text{Total Liabilities}$	> 1.5x
Debt-to-Equity	$\text{Total Liabilities} / \text{Equity}$	< 2.0x
Cash Balance	Bank account total from balance sheet	> \$10K
Debt-to-Asset %	$\text{Total Liabilities} / \text{Total Assets} \times 100$	< 80%
Equity Ratio %	$\text{Equity} / \text{Total Assets} \times 100$	> 20%
Net Debt	Long-term Loans - Cash	Monitor
DSCR (Est.)	$\text{NOI} / (\text{Interest Expense} \times 1.2)$	> 1.25x

Occupancy KPIs (when ops data is linked)

Card Label	Formula	Target
Occupancy Rate	From registry occupancy %	> 95%
Economic Occupancy / Rent Collection Rate	Collected / Billed x 100	> 95%
Vacancy Rate	100 - Occupancy Rate	< 5%
Cap Rate	NOI / Buildings x 100	> 5%
Revenue per Unit	Total Revenue / Total Units	Trend
Expense per Unit	Total Expenses / Total Units	Trend

C) CFO Dashboard (Financial Overview - P&L charts)

Data source: first company's uploaded financials Route: Rentals -> CFO Dashboard Frontend reference: frontend/src/pages/RentalCfoDashboard.tsx

Metric / Chart	Formula		
Revenue	Total Revenue or sum of revenue/rental income lines		
Expenses	Sum of expense lines		
Net Income	Net Income row		
Net Margin	Net Income / Revenue x 100		
Expense Ratio Trend	Expenses / Revenue x 100 by year		
Cash (Bank)	Total Bank Accounts from balance sheet		
Net Income Trajectory	Net Income by year		
Latest Net Income change	$\frac{\text{Latest NI} - \text{Prior NI}}{\text{Prior NI}}$	Prior NI \	x 100`
Avg Profit Margin	Average of yearly (Net Income / Revenue x 100)		

D) CFO Portfolio (strategic portfolio view)

Data source: companies, expenses, units, loans, portfolio summary Route: Rentals -> CFO Portfolio Frontend reference: frontend/src/pages/rental/RentalCfoPortfolio.tsx, frontend/src/hooks/useRentalCfoData.ts

Top KPI tiles

Card Label	Formula
NOI	Collected This Month - Total Expenses (filtered companies)
Occupancy	Occupied Units / Total Units
Collection Rate	Collected / Billed
Expense Ratio	Total Expenses / Gross Potential Rent
DSCR	$(\text{NOI} \times 12) / (\text{Total Monthly EMI} \times 12)$
Cash Position	NOI - Total Monthly EMI
Rent Growth	Currently static placeholder (+2.4%)
Vacancy Cost	Portfolio vacancy_loss

Portfolio Health Score (out of 100)

Component	Max Points	Formula
Occupancy	25	$\min(25, \text{Occupancy \%} \times 25)$
Collections	25	$\min(25, \text{Collection Rate} \times 25)$; if no billing data -> 15
Expense Control	25	25 if expense ratio < 30%; 18 if < 45%; else 10
Debt Coverage	25	25 if DSCR > 1.25; 18 if >= 1.0; 12 if no loan data; else 8

Total Health Score = sum of the 4 components (averaged across filtered companies)

Company Comparison Matrix

Column	Formula
Occupancy	Company occupancy %

Column	Formula
NOI	Collected - Expenses for the month
Exp Ratio	Expenses / Gross Potential Rent
DSCR	(NOI x 12) / (Company EMI x 12)
Cash	NOI - Company EMI
Score	Same health score formula above
Flag	Score < 70 or DSCR < 1.0

Building-level (Building Expenses view)

Metric	Formula
Rent Income	Collected (or GPR if no collection)
Total Expenses	Sum of building expenses for current month
NOI	Rent Income - Total Expenses
Expense Ratio	Total Expenses / Rent Income
NOI Margin	NOI / Rent Income
Status	Healthy if ratio < 30%; Watch if <= 45%; High if > 45%

G) Financial Ratios page

Route: Rentals -> Financial Ratios Frontend reference: frontend/src/pages/rental/RentalFinancialRatios.tsx

Profitability ratios

Ratio	Formula	Benchmark
NOI Margin	NOI / Revenue x 100	> 35%
Net Profit Margin	Net Income / Revenue x 100	> 10%
Operating Expense Ratio	Total OpEx / Revenue x 100	< 60%
EBITDA Margin	EBITDA / Revenue x 100	> 45%
Return on Assets (ROA)	Net Income / Total Assets x 100	> 4%
Return on Equity (ROE)	Net Income / Equity x 100	> 8%
Gross Rent Multiple	Asset Value / Annual Revenue	< 14x
Cash-on-Cash Return	Pre-tax Cash Flow / Equity x 100	> 7%

Where: - NOI = Revenue - Expenses + Interest Expense - EBITDA = NOI + Depreciation + Amortization

Liquidity ratios

Ratio	Formula	Benchmark
Current Ratio	Current Assets / Current Liabilities	> 1.5x
Quick Ratio	(Current Assets - Inventory) / Current Liabilities	> 1.0x
Cash Ratio	Cash & Bank / Current Liabilities	> 0.2x
Operating CF Ratio	Operating Cash Flow / Current Liabilities	> 1.0x
Working Capital	Current Assets - Current Liabilities	Positive
Days Cash on Hand	Cash / (Annual OpEx / 365)	> 60 days

Solvency / leverage ratios

Ratio	Formula	Benchmark
Debt-to-Equity	Total Liabilities / Equity	< 5x (RE)
Debt-to-Asset	Total Liabilities / Total Assets x 100	< 80%
Equity Ratio	Equity / Total Assets x 100	> 20%
Interest Coverage	NOI / Interest Expense	> 1.5x
LTV	Mortgage / Long-term Loans / Property Value (Buildings) x 100	< 80%
Net Debt	Long-term Loans - Cash	Monitor
DSCR	NOI / Total Debt Service (est: NOI / Interest x 1.2)	> 1.25x
Debt Service Ratio	Debt Service / NOI x 100	< 65%
Fixed Charge Coverage	(NOI + Fixed Charges) / Fixed Charges	> 1.25x

Occupancy & pricing ratios

Ratio	Formula	Benchmark
Occupancy Rate	Occupied Units / Total Units x 100	> 90%
Economic Occupancy	Rent Collected / Gross Potential x 100	> 92%
Rent Collection Rate	Collected / Billed x 100	> 95%
Vacancy Rate	Vacant Units / Total Units x 100	< 10%
Loss to Lease	(Market Rent - Actual Rent) / Market Rent x 100	< 5%
Avg Days Vacant	Average days between tenants	< 21–30 days
Rent per Sq Ft	Average Rent / Average Sq Ft	Market
Revenue per Unit	Total Revenue / Total Units	Trend
Expense per Unit	Total OpEx / Occupied Units	< \$1,000/mo
Cap Rate	NOI / Property Value x 100	4–6% (market)
Price / Rent Ratio	Property Value / Annual Rent	< 20x
EGIM	Property Value / Effective Gross Income	< 12x

Returns / cost of capital

Ratio	Formula
WACC	Weighted average cost of debt and equity
Cost of Debt	Interest Expense / Total Debt x 100
Cost of Equity	Required return on equity (CAPM estimate)
Return vs WACC	Portfolio Return - WACC
Spread (Cap - WACC)	Cap Rate - WACC

H) AR Dashboard

Route: Rentals -> AR Dashboard Frontend reference: frontend/src/pages/RentalArDashboard.tsx

KPI	Formula
Total Billed / Month	Sum of billed rent from registry (occupied units)
Collected (Latest Mo)	Sum of Rent Receivable collections for selected month
Outstanding AR	Billed - Collected (open balance)
Collection Rate	Collected / Billed x 100
Vacancy Loss / Month	Sum of rent for vacant/notice units from registry
Zero-Pay Companies	Companies with billed > 0 and collected = 0
Partial-Pay Companies	0 < Collected < Billed
Month-End Shortfall	Billed - Collected for selected month
Occupied - Billing Gap	Occupied units with vs without billing data
Top 5 by Outstanding AR	Companies ranked by highest open AR
DSO (Arrears Days)	Weighted bucket formula (same as Overview)

Collection detail table (per company x month): - Billed = registry occupied-unit rent - Collected = Rent Receivable upload (or P&L fallback)
- Outstanding = Billed - Collected - Rate = Collected / Billed x 100 - Status: Zero-Pay (0%), Partial (<100%), Paid (>=95%), Low (<85%)

I) Quick reference - core definitions

Occupancy % = Occupied Units / Total Units
Collection Rate = Collected / Billed
NOI (Ops) = Collected - Expenses
NOI (Financial) = Revenue - Expenses + Interest
Vacancy Loss = Rent from vacant units only
Arrears = Sum of (Billed - Collected), minimum 0
DSCR = Annual NOI / Annual Debt Service (EMI x 12)

J) Code locations

Area	Primary files
Operational KPIs (backend)	backend/services/rental_calculations.py, backend/routers/rentals/router.py
Overview UI	frontend/src/pages/RentalOverview.tsx
Financial KPI engine	frontend/src/utis/rentalKpiEngine.ts
Financials KPI tab	frontend/src/pages/RentalFinancials.tsx

Area	Primary files
CFO Dashboard	frontend/src/pages/RentalCfoDashboard.tsx
CFO Portfolio	frontend/src/pages/rental/RentalCfoPortfolio.tsx
CFO data hook	frontend/src/hooks/useRentalCfoData.ts
Financial Ratios	frontend/src/pages/rental/RentalFinancialRatios.tsx
Executive Summary / PPT export	frontend/src/utills/executiveSummaryPpt.ts, frontend/src/utills/gatherExecutiveExportData.ts

Last updated: July 2026 - CSV companion: docs/Rentals-KPI-Formulas.csv